# Ramchander Rao Bhaskara

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## Appointments

**Dartmouth College** — Dec 2025 – Present
Department of Computer Science
*Postdoctoral Associate* | *Advised by Prof. Adithya Pediredla*

**Texas A&M University** — Sep – Dec 2025
Department of Aerospace Engineering
*Research Engineer*

## Education

**Texas A&M University** — 2021 – 2025
PhD in Aerospace Engineering | *Advised by Prof. Manoranjan Majji*

**Texas A&M University** — 2019 – 2021
M.S. in Aerospace Engineering | *Advised by Prof. Manoranjan Majji*

**National Institute of Technology, Trichy** — 2013 – 2017
BTech in Instrumentation and Control Engineering

## Awards

**Best Talk Award** — 2026
Postdoctoral Research Day, Dartmouth College and University of New Hampshire

**Excellent Reviewer** — 2025
Journal of Guidance, Control, and Dynamics (JGCD)

**Guidance, Navigation, and Control (GNC) Graduate Award** — 2024
American Institute of Aeronautics and Astronautics | *3,500 USD*

**Graduate Excellence Fellowship** — 2020 – 2024
Texas A&M University | *5 × 1,000 USD*

**Travel Award** — 2023, 2024
Department of Aerospace Engineering, Texas A&M University | *500 USD*

**Finalist, GNC Graduate Student Paper Award** — 2023
AIAA SciTech Forum | *500 USD*

**Best Student Research Paper Award** — 2022
Digital Avionics Systems Conference (DASC) | *500 USD*

**Travel Grant** — 2022
Office of Graduate and Professional Studies, Texas A&M University | *500 USD*

**ASIE Scholarship** — 2022
American Society of Indian Engineers and Architects, Houston | *1,000 USD*

**NASA TechLeap Prize — GNC Lead** — 2021
NASA Flight Opportunities Program | *100,000 USD towards flight experiment*

**RECT Silver 72 Scholarship** — 2015 – 2017
National Institute of Technology, Trichy | *15,000 INR per year*

**IIT Madras Summer Research Fellowship** — 2016
Department of Aerospace Engineering, IIT Madras | *Granted but declined*

## Other Research Experience

**Cornell University** 2025
Mechanical and Aerospace Engineering | *SmallSat Mission Design School*

**NASA Jet Propulsion Laboratory, Caltech** 2022, 2023
Mobility and Robotic Systems Section | *Research Intern*

**iRunway India (now UnitedLex)** 2017 – 2019
Intellectual Property | *Associate (5G/LTE network architecture)*

## Publications

**Development and Validation of Velocimeter Lidar Simulator**
**Ramchander Rao Bhaskara**, Roshan T. Eapen, and Manoranjan Majji
*AIAA SciTech 2025*

**Searching for Ariel's Aliens with the CRISPI Mass Spectrometer SmallSat Flyby Mission**
Zach Ulibarri, S Jamal, A Alex, L Arzoumanian, **Ramchander Rao Bhaskara**, A Chadwick, S Gayen et al.
*American Geophysical Union 2025*

**Quantized State Estimation for Linear Dynamical Systems**
**Ramchander Rao Bhaskara**, Manoranjan Majji, and Felipe Guzman
*Sensors 2024*

**Icy Moon Surface Simulation and Stereo Depth Estimation for Sampling Autonomy**
**Ramchander Rao Bhaskara**, G Georgakis, J Nash, J Bowkett, M Cameron, A Ansar, P Backes, and M Majji
*IEEE Aerospace Conference 2024*

**Differentiable Rendering for Pose Estimation in Proximity Operations**
**Ramchander Rao Bhaskara**, Roshan T. Eapen, and Manoranjan Majji
*AIAA SciTech Forum 2023* | *Finalist, GNC graduate student papers*

**An FPGA Framework for Interferometric Vision-Based Navigation (iVisNav)**
**Ramchander Rao Bhaskara**, Kookjin Sung, and Manoranjan Majji
$41^{st}$ *Digital Avionics and Systems Conference 2022* | *Best student research paper*

**FPGA Hardware Acceleration for Feature-Based Relative Navigation Applications**
**Ramchander Rao Bhaskara** and Manoranjan Majji
*AAS/AIAA Astrodynamics Specialist Conference 2022*

**Vision and Inertial Sensor Fusion for Terrain Relative Navigation**
A. Verras, R. Eapen, A Simon, M Majji, **Ramchander Rao Bhaskara**, C. Restrepo, and R. Lovelace
*AIAA SciTech Forum 2021*

**Interferometric Vision-Based Navigation Sensor for Autonomous Proximity Operation**
Kookjin Sung, **Ramchander Rao Bhaskara**, and Manoranjan Majji
$39^{th}$ *Digital Avionics and Systems Conference 2020*

### *Under Review*

A Paper on Laser-Based Depth Scanning Instrument
**Ramchander Rao Bhaskara**, Dhawal Sirikonda, Omkar Vengurlekar, Juhyeon Kim, Suren Jayasuriya, Adithya Pediredla

A Paper on Event-Based Sensors for Low-Bandwidth Surveillance
Nevindu Batagoda, **Ramchander Rao Bhaskara**, Christian Metzler, Adithya Pediredla

On the Low-Pass Filter Design for Digital Phasemeter
**Ramchander Rao Bhaskara** and Manoranjan Majji

## Other Publications

**Real-Time Signal Processing and State Estimation for Spaceflight Applications**
**Ramchander Rao Bhaskara**
*PhD Dissertation, supervised by Prof. Manoranjan Majji*

**Hardware Implementation of Navigation Filters for Automation of Dynamical Systems**
**Ramchander Rao Bhaskara**
*Master's Thesis, supervised by Prof. Manoranjan Majji*

**Precision Depth Sensing: Space Science to Robotics**
**Ramchander Rao Bhaskara**
*Guarini Poster Session, Dartmouth College, 2026*

**NaRPA: Navigation and Rendering Pipeline for Astronautics**
Roshan T. Eapen*, **Ramchander Rao Bhaskara*** and Manoranjan Majji
*Poster, Lunar Surface Innovation Consortium 2021*

**Physics-Based Modeling of Selective Catalytic Reduction System and Reduced Order Dynamics**
**Ramchander Rao Bhaskara**
*Undergraduate Thesis, supervised by Prof. Umapathy Mangalanathan*

## Talks

**Optimal Estimation Under Uncertainty**
Visual Computing Seminar | *hosted by Prof. Adithya Pediredla* — Dartmouth College | *2026*

**On-Board Sensing and State Estimation for Opto-Mechanical Inertial Sensors**
Starlink (SpaceX) | *hosted by Javier Vicens (GNC)* — Seattle | *2025*

**Computer Vision for Icy-World Exploration**
Machine-Ground Interaction Consortium (MaGIC) | *Invited Talk* — University of Wisconsin-Madison | *2025*

**Optical Instrumentation for Space Science and Robotics**
Postdoctoral Research Day — Dartmouth College | *2026*
Visual Computing Seminar | *Invited Talk* — Dartmouth College | *2025*

**Sampling Autonomy for Europa Lander**
Jet Propulsion Laboratory | *Intern Talk* — Caltech | *2023*

**Study of Topology of Icy Moons**
Jet Propulsion Laboratory | *Intern Talk* — Caltech | *2023*

**Texas A&M ScORE: Space Object Rendering Engine**
Lunar Surface Innovation Consortium | *Symposium* — Johns Hopkins University | *2021*

## Funding

**Sensing for Climate-Resilient Mobility: Enabling the Energy Transition** *2026*
Irving Seed Grant and Pilot Funding Program | *Co-PI: Prof. Adithya Pediredla*
*Selected, final phase*

**Developing Sensor Systems for Adverse Weather Conditions** *2026*
Neukom Postdoctoral Fellowship, Dartmouth College
*Finalist, Not selected*

**Event-Based Sensors for Meteoroid Tracking** *2025*
NASA Postdoctoral Program | *Co-PI: Paul Abell (NASA JSC)*
*Not selected*

**High-Fidelity Terrain and Sensor Simulation Pipeline** *2024*
JPL Strategic University Research Program | *PIs: Manoranjan Majji & Georgios Georgakis (NASA JPL)*
*Not selected*

## Academic Service

**American Institute of Aeronautics and Astronautics (AIAA)** *2024 – Present*
Sensor Systems and Information Fusion (SSIF) | *Technical Committee*

**AIAA SciTech Forum** *2025*
Guidance, Navigation, and Control (GNC-44) | *Session Chair*

**AIAA SciTech Forum** *2025*
Sensor and Algorithm Analysis (SEN-02) | *Session Chair*

## Reviews

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|---|---|
| Journal of Guidance, Control, and Dynamics (JGCD) | 2025 |
| Journal of Astronautical Sciences (JAS) | 2024, 2025 |
| IEEE Control Systems Letters | 2025 |
| American Control Conference | 2022 – 2025 |
| AIAA SciTech Forum | 2025 |
| Transactions on Computers | 2023 |

## Teaching

**AERO 423: Orbital Mechanics** *2024, 2025*
Texas A&M University | *Teaching Assistant*

**Digital Signal Processing** *2023*
Texas A&M University | *Department Tutorial Lecture*

## Mentoring

**Dhawal Sirikonda** *2026*
Dartmouth College | *PhD research in digital signal processing*

**Nevindu Batagoda** *2026*
Dartmouth College | *PhD research in image signal processing*

**Piren Li** *2026*
Dartmouth College | *MS research in optics*

**Omkar Vengurlekar** *2026*
Arizona State University | *Graduate research in audio signal processing*

**Team Lunatyx** *2025*
PI for Lunar Autonomy Challenge 2025 | *2 graduate and 3 undergraduate students, TAMU*

**Omar Mohmand & Marco Peredo** *2024*
Texas A&M University | *Undergraduate research: Trajectory design for Mars rendezvous*

## Volunteering

**Montshire Museum of Science** *2026*
Astronomy Day | *Speaker, Astronomer in the House*

**Aerospace Engineering Graduate Student Association (AEGSA)** *2023 – 2025*
Texas A&M University | *Professional Development Chair*

**Texas A&M University** *2022 – 2025*
Science and Astronomy Festival | *Volunteer, Outreach Demonstrations*

**Bhumi NGO** *2017 – 2019*
Renuka High School, Bangalore | *Volunteer Physics Teacher*

## Software

**GUISS**: Graphical Utility for Icy Moon Surface Simulations 2024